

Science Olympiad — Cryptalicious Race - Round 4

Names of participants: (Please print neatly)

School Name:

Team Number:

Check your team: ☐ Varsity ☐ JV1 ☐ JV2 ☐ JV3

Warning: Do not open this packet until given permission to do so.

Note: There are useful notes after this page.

Question	Value	Incorrect letters	Deduction	Score
1	0			
2	0			
3	0		≡	
4	0			
Final Score				

1) **[0 points]** You find an encoded message in the test tube rack, written on masking tape. It's a quote, written in a Homophonic. The deciphered text ends with **ISID**. What does it say?

23 67 73 51 00 87 47 60 70 69 64 63 6 72 76 11 60 24 26 99

60 10 1 4 39 53 26 43 35 2 63 3 73 14 5 68 43 6 29

2) **[0 points]** The message below was intercepted from an experimental satellite transmission. It contains a quote, encoded as a Caesar. What does it say?

X	J	H	T	S	I	D	R	P	A	A	Q	P	R	Z	H	R	P	B	B	T	G	H	P	C	S
L	P	H	I	T	I	W	T	X	G	I	X	B	T												

3) **[0 points]** A coded message is discovered inside a fortune cookie. It turns out to be a quote, encoded using a Cryptarithm Cipher. What do the values 9023 871 465 decode to? RECALL THE TRICK WE LEARNED IN CLASS TO CONVERT SUBTRACTION CRYPTARITHMS INTO AN EASIER PROBLEM...

Values to decode for solution

9	0	2	3	8	7	1	4	6	5

Cryptarithm formula

D I N G H Y

- K A Y A K

-

S H I P

	0	1	2	3	4	5	6	7	8	9
A										
D										
G										
H										
I										
K										
N										
P										
S										
Y										

4) **[0 points]** A message in a bottle washed up in the school's decorative fountain—decode this Complete Columnar cipher text. The quote has **BEACHIS** somewhere in it. What does it say?

NOGHB BSERS EURAM EDAIH XOVNT SYNFT OEYTD YTOCN

EXAIT IMIXO ITBRO NFHAH GRX

Answer: _____